Angelo Guarnerio

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EDUCATION

Politecnico di Milano — Milan, Italy
MSc in Biomedical Engineering (GPA: 28.4/30) — Expected graduation: 2027

Relevant Coursework Projects

- **Embedded and Edge AI:** on-device audio scene recognition for smartglasses with **Luxottica**; spectrogram and time-domain features, deployed with **TensorFlow Lite** and **Edge Impulse**.
- **Artificial Neural Networks and Deep Learning: LSTMs** for pain detection from time series and **Vision Transformers (ViT)** for tumor classification in medical images.
- **Creativity, Science and Innovation:** *Zenit*, a stress-aware smart calendar; built the deep-learning pipeline forecasting stress from Garmin wearable data. **3rd place** in a **Bending Spoons** competition.

RESEARCH EXPERIENCE

IDUN Technologies AG – ETH Zurich Spin-off — Oct 2026 – Present
Data Science Intern, Master Thesis — Zürich, Switzerland

- **Research:** master thesis on physiological signals, primarily **in-ear EEG**, to derive **cognitive metrics**.
- **Study design:** designed the experimental study, defining the protocol and the data-collection procedure.
- **Methods: signal processing** for EEG feature extraction, **statistical learning** for modelling and validation.

NECST Laboratory – Politecnico di Milano — Feb 2024 – Present
Student Researcher — Milan, Italy

- **Stress Detection:** developed a novel mental stress index by fine-tuning **NormWear**, a **foundation model** for wearable physiological time series, on laboratory data in **PyTorch**. Implemented a regression head that predicts stress levels from validated psychometric questionnaires, with the architecture designed for **federated learning** so that raw data never leaves the user's device.
- **Time-Series Patient Clustering:** learned compact representations of high-dimensional ICU physiological time series from the **MIMIC-IV** database with **Variational Autoencoders** (PyTorch), then clustered the embeddings (**scikit-learn**) to identify distinct clinical phenotypes.
- **BACCO, Holistic Wellness Monitoring:** developed an **iOS app** (**Swift**, HealthKit) for daily self-assessment, combining **HRV metrics** (SDNN, RMSSD) with behavioral data on physical activity and diet into a single daily wellness index.

PUBLICATIONS

- L. Ginestretti, S. Bardini, **A. Guarnerio**, S. Varone, *Embedding Based Learning for Stress Assessment from Wearable Time Series*, IEEE EMBC 2026.
- **A. Guarnerio**, L. Fiorin, A. Isella, *BACCO: a Tool for Holistic Wellness Monitoring*, IEEE STAR 2024.

HACKATHONS

- **START Hack 2026, Top 3** (St. Gallen, one of Europe's largest): built a shared workspace powered by **AI agents** for students and supervisors to streamline the thesis journey.
- **Humanitas AI Hackathon 2026:** clinician dashboard for wearable monitoring of rheumatoid-arthritis patients: accelerometer-derived metrics, alerts and AI-generated clinical summaries.

ASSOCIATIONS AND LANGUAGES

- **Entrepreneurship Club Polimi:** Information Technology member and workshop organizer.
- **CoderDojo Milano:** volunteer coding mentor for children in hands-on programming workshops.
- **Lead The Future Mentorship:** one of the few Italian students selected as mentees by this STEM mentorship non-profit.
- **Languages:** Italian (native), English (C1, TOEIC).

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